ALMA MATER STUDIORUM • UNIVERSITÀ DI BOLOGNA
DIPARTIMENTO DI INFORMATICA – SCIENZA E INGEGNERIA (DISI)

OCI Image Registry Continuum:

Performance Characterization & Resource Overhead

Infrastructures for Cloud Computing and Big Data M

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The Cloud-to-Edge Continuum & OCI Image Registries

- **Modern Deployment Paradigms:** Shift from static, environment-dependent configuration to packed, immutable containerized shipping (Docker, Kubernetes).
- **The Role of Registries:** Container image registries have evolved from passive, out-of-line storage bins into critical, runtime-active middleware components.
- **The Geodistributed Bottleneck:** Multi-region orchestration introduces severe performance challenges, governed by physical network propagation delays (RTT), Bandwidth-Delay Product (BDP) limits, and transactional consensus barriers (CAP).

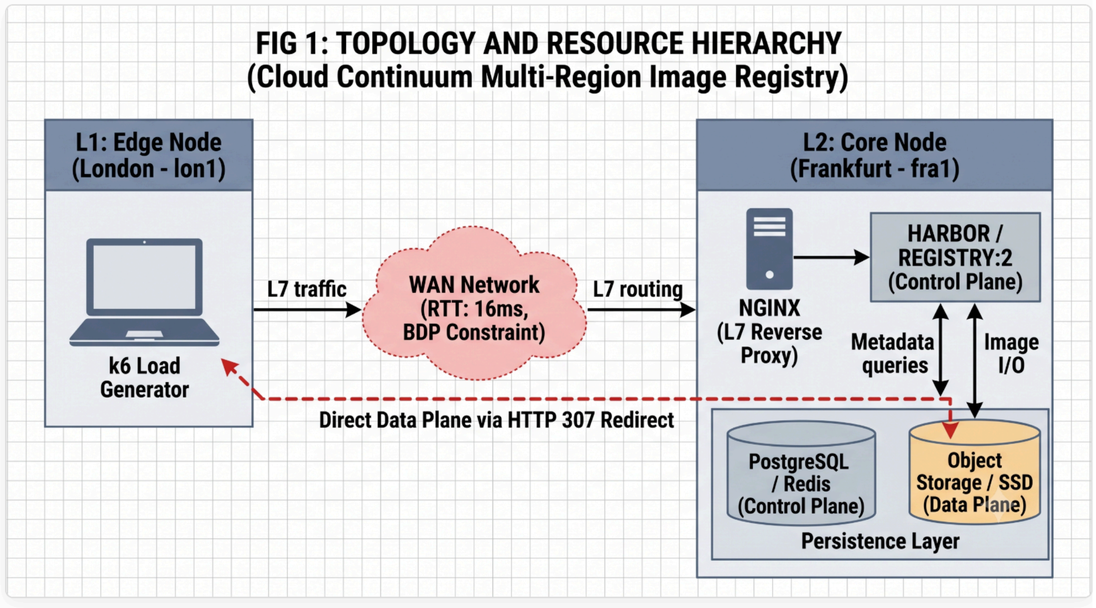


Fig. 1: Control vs. Data Plane decoupling in geodistributed OCI Registry topologies.

Architectural Trade-offs: Transactional ACID vs. Elastic BASE

- **Stateless Registry (ACID-oriented):** Direct, POSIX-compliant writes to local block storage (Registry:2 on Local SSD). Prioritizes immediate strong consistency, transactional safety, and low single-user latency, but remains highly vulnerable to "data gravity" bottlenecks under high concurrent access.
- **Stateful Registry (BASE-oriented):** Complex microservice ecosystem (Harbor on S3 Spaces) coupling a relational control plane (PostgreSQL/Redis) with decoupled cloud object storage. Embraces eventual consistency and L7 redirects to enable horizontal, geographical elasticity.
- **The Core Question:** How does L7 Control Plane complexity (Statefulness) impact runtime performance and resource efficiency when compared to physical, Layer 4 transport-layer constraints?

Metric / Property	ACID (Registry:2)	BASE (Harbor)
Consistency Model	Strong (Immediate POSIX)	Eventual (Replication Lag)
Persistence Layer	Local SSD (Block)	S3 Spaces (Object)
Scaling Strategy	Vertical (Host Bound)	Horizontal (Cloud Elastic)
Concurrency Path	Pessimistic (Locks/Queues)	Optimistic (L7 Offloading)
Coordination Tax (T_0)	Minimal ($T_0 \approx 0$)	High Static & Dynamic Tax

Experimental Setup & Geodistributed Testbed

- **Infrastructure Nodes:**
  - **Core Server (Frankfurt - fra1):** Deployed on a DigitalOcean VM (2 vCPUs, 4 GB RAM) running Ubuntu 24.04 LTS. Hosts the Registry:2 stateless engine, Harbor stateful control plane, and NGINX L7 Reverse Proxy.
  - **Edge Client (London - lon1):** Deployed on an isolated DigitalOcean VM (1 vCPU, 2 GB RAM) dedicated to distributed load generation via Grafana k6.
- **Cloud Persistence:** S3-compatible Object Storage (DigitalOcean Spaces) located in Frankfurt, integrated with the Harbor registry path.
- **Workload Profiles (Grafana k6):**
  - **Script A (Metadata-Heavy):** High frequency, low payload GET Manifest queries to stress L7 governance and connection pools.
  - **Script B (Payload-Heavy):** High-volume, concurrent multi-megabyte OCI image layer transfers to saturate transport and I/O paths.

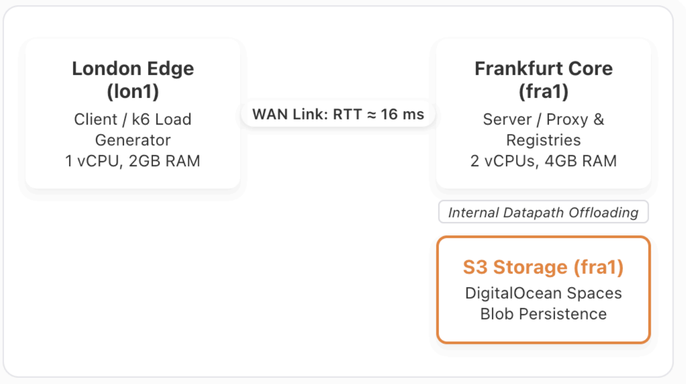
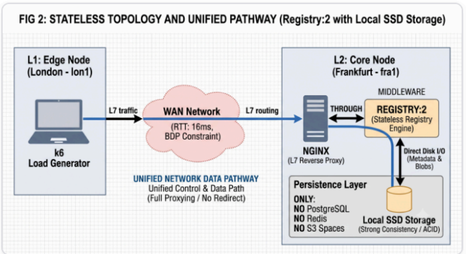


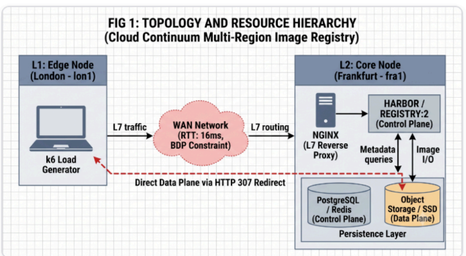
Fig. 3: Geodistributed experimental topology and physical node specifications.

Control Plane vs. Data Plane: Schematic Comparison

- **Registry:2 Unified Network Path:**
  - Operates via **Synchronous L7 Proxying**. NGINX actively intercepts, terminates TLS, parses, and proxies *all* control traffic (manifests) and heavy data traffic (payload layers) directly to the local SSD persistence layer.
  - The reverse proxy and the registry engine share the critical path, making the host highly susceptible to local I/O queue starvation under concurrency.
- **Harbor Decoupled Datapath:**
  - Employs **L7 Data Plane Offloading**. The L7 Control Plane (Harbor Core) validates client credentials against PostgreSQL, then instantly offloads the heavy binary I/O stream to S3 Object Storage via **HTTP 307 Temporary Redirects**.
  - Bypasses the NGINX reverse proxy entirely during the binary streaming phase, protecting the host core from network bufferbloat.
- **The "Sidecar Tax" Hypothesis:** Stateful, multi-container synchronization (PostgreSQL, Redis, Core) is predicted to introduce a measurable latency and CPU coordination overhead ($T_0(N)$) during metadata-heavy operations.



REGISTRY:2 (UNIFIED PATH)



HARBOR (DECOUPLED PATH VIA HTTP 307)

Fig. 4: Architectural comparison of the unified synchronous storage pathway vs. decoupled control/data plane offloading.

L7 Metadata Control Plane: Quantifying the "Sidecar Tax"

- **Formal Efficiency Model (Amdahl's Law):**
  We formalize the middleware tax using the distributed systems efficiency model:

$$E_P = 1 / (1 + T_0(N) / T_1(N))$$

  where **$T_1(N)$** represents useful execution work, and **$T_0(N)$** represents the non-parallelizable coordination and synchronization overhead.

- **Registry:2 Efficiency ($T_0 \approx 0$):**
  Achieves a median metadata latency of **22.08 ms**. Since the stateless engine avoids relational database and loopback overhead, the coordination tax is virtually non-existent, pushing efficiency near-ideal ($E_P \approx 1$) and operating close to the physical WAN RTT floor.

- **Harbor Stateful Penalty (The 'Sidecar Tax'):**
  Median metadata latency jumps to **80.56 ms**. Serving the same L7 GET Manifest request requires an additional ~58 ms of compute time spent in PostgreSQL transaction validation, Redis session cache lookup, and OIDC authorization.

- **Tail Latency & Outliers:**
  High-concurrency metadata stress (500 VUs) triggers PostgreSQL Connection Pool Starvation, driving queuing delays exponentially up to the 30s SLA limit.

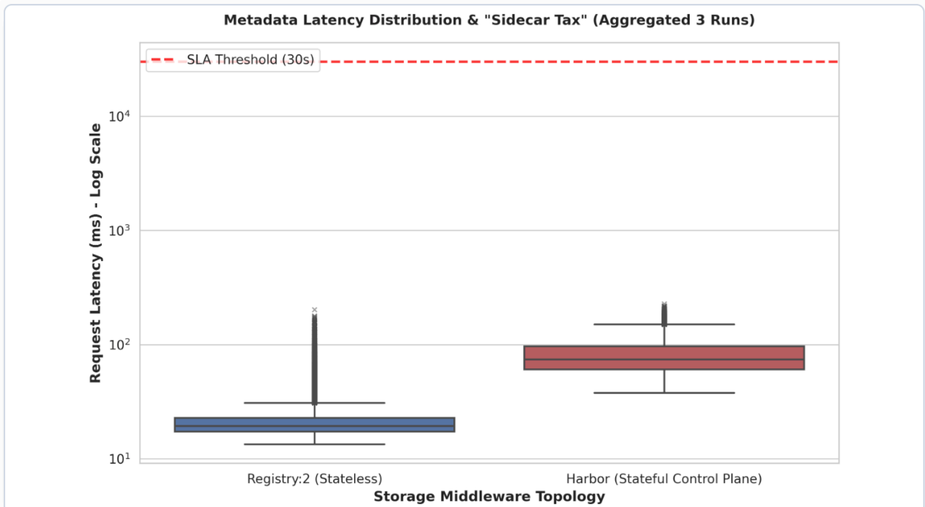


Fig. 5: Empirical latency distribution under 500 VUs metadata stress (logarithmic scale), isolating the static and dynamic L7 coordination tax of Harbor.

Control Plane Saturation & Right-Censoring Mitigation

- **PostgreSQL Connection Starvation:**
  Sustained peak metadata concurrency (500 VUs) saturates Harbor's multi-container control plane, quickly depleting the database connection pool and driving queuing delays ($\mathbf{T_q}$) exponentially.
- **Systematic SLA Breaches:**
  Cumulative queue delay rapidly exceeds the 30-second application threshold, forcing the k6 load generator to drop active TCP connections and record HTTP gateway timeouts.
- **Right-Censoring Mitigation:**
  To prevent optimistic bias (where 30s timeouts artificially compress mean latencies), uncompleted samples are mathematically isolated, preserving the true asymptotic scale of the control-plane collapse.
- **Self-Induced Thrashing Cycle:**
  Sudden queue truncation from timeouts temporarily relieves database pressure (dropping CPU and throughput), only to re-saturate upon client reconnection in a destructive, cyclical collapse.

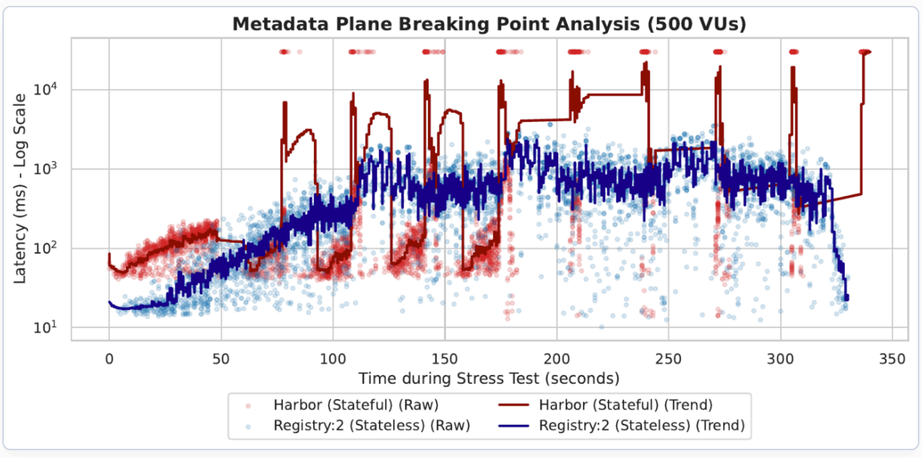


Fig. 6: Metadata breaking point: Latency evolution under 500 VUs, illustrating the transition from linear scaling to exponential database queue collapse.

L7 Data Plane: Throughput Scalability (1 - 100 VUs)

- **Linear Scaling Zone (≤ 40 VUs):**
  Both Registry:2 and Harbor scale identically and linearly, climbing up to ~150 MiB/s. This empirical overlap demonstrates the complete success of Harbor's **Datapath Decoupling** strategy: under moderate load, the stateful control plane does not penalize binary payload operations.
- **Physical Saturation Plateau (> 40 VUs):**
  Beyond 40 VUs, the throughput curves flatten, converging and fluctuating within a narrow band of **200–240 MiB/s**. The expansion of variance bands (shaded regions) flags the onset of physical WAN congestion, transport packet loss, and buffer jitter.
- **Capped Scale (100 VUs Limit) - A Methodological Choice:**
  Extending the concurrency axis up to 500 VUs on this plot would be scientifically redundant. Because the geodistributed WAN link is physically saturated, higher client pressure would only yield a degraded, flat plateau with TCP window throttling, offering no new architectural insights.

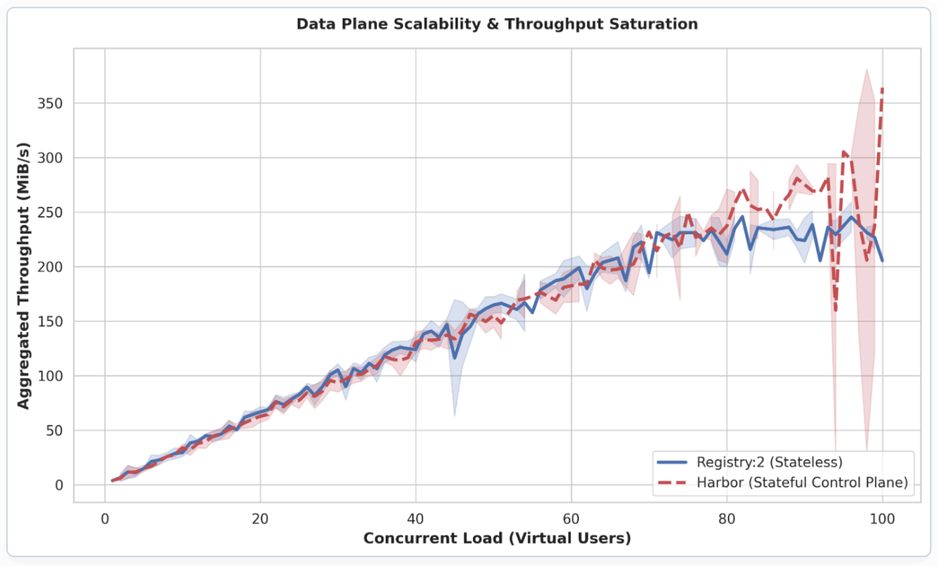


Fig. 7: L7 aggregate heavy payload throughput scaling. Beyond 40 VUs, both registry pipelines converge to the physical saturation plateau of the network link.

Payload Breaking Point: BASE Offloading vs. ACID Data Gravity

- **The Local SSD Bottleneck (ACID Data Gravity):**
  Under a maximum concurrency of 500 VUs, Registry:2 (writing directly to local SSD) suffers from severe write-queue starvation on the host's storage controller. The POSIX-compliant synchronous ACID path creates an I/O bottleneck, driving average write latencies up to **10 seconds** and triggering systematic 30-second timeouts.
- **Harbor S3 Offloading Stability (BASE Paradigm):**
  Harbor utilizes L7 Data Plane Offloading via HTTP 307 Redirects. Once the client is authenticated, Harbor completely delegates the binary data stream directly to DigitalOcean Spaces (S3 Object Storage).
- **Stabilized Response Times:**
  By bypassing the host proxy and disk bottlenecks entirely, Harbor's payload write latencies stabilize on a flat, compact plateau between **4 and 6 seconds**, with zero cascading queue failures or systematic SLA timeouts.
- **Architectural Outcome:**
  Highly parallel, eventually consistent distributed storage (BASE) successfully bypasses core host I/O constraints, demonstrating superior scalability for data planes in the geodistributed Cloud Continuum.

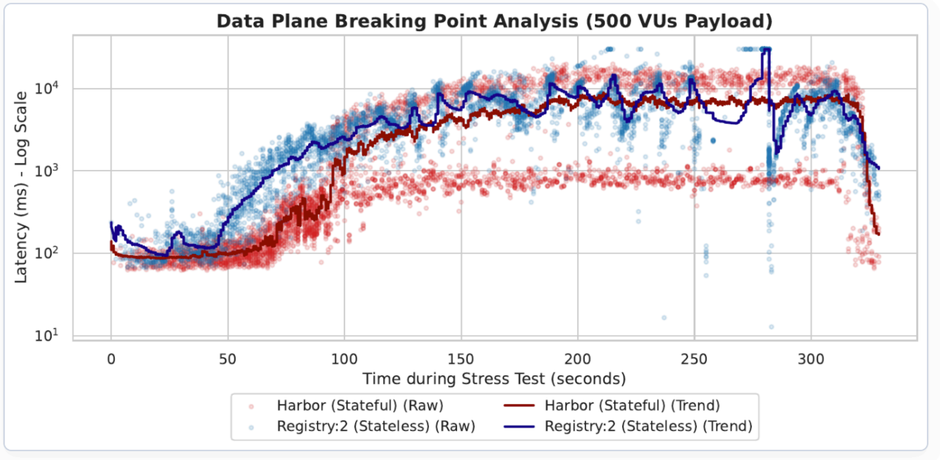


Fig. 8: Heavy payload breaking point: Latency evolution under 500 VUs. S3 offloading (BASE) prevents the local host write-queue starvation suffered by Registry:2 (ACID).

L4 Transport Layer Limits: Cumulative BDP Analysis

- **The Bandwidth-Delay Product (BDP) Model:**
  The fundamental data capacity of a network pipe is governed by transport physics:

$$BDP = Capacity \times RTT$$

- **Single-Socket Throughput Boundary:**
  For any individual unconstrained TCP connection, the maximum achievable throughput is bounded by the receive window size ($\mathbf{W_{max}}$) and the Round-Trip Time (RTT):

$$Throughput_{max} \leq W_{max} / RTT$$

- **Constrained Buffer Regime:**
  By forcing the kernel TCP window to a fixed $\mathbf{W_{max}}$ **= 64 KB** over our geodistributed WAN link (**RTT ≈ 16 ms**), the single-socket ceiling is strictly capped at:

$$Throughput_{max} \leq 64\ KB / 16\ ms \approx 4.09\ MB/s$$

- **Empirical Validation of L4 Domination:**
  Under high-concurrency multi-socket stress (k6), cumulative data transfer (red curve) perfectly matches the predicted theoretical aggregate slope of **~54.0 MB/s** (dashed black line) due to **sender stalling**, proving network physics override application efficiency under buffer constraints.

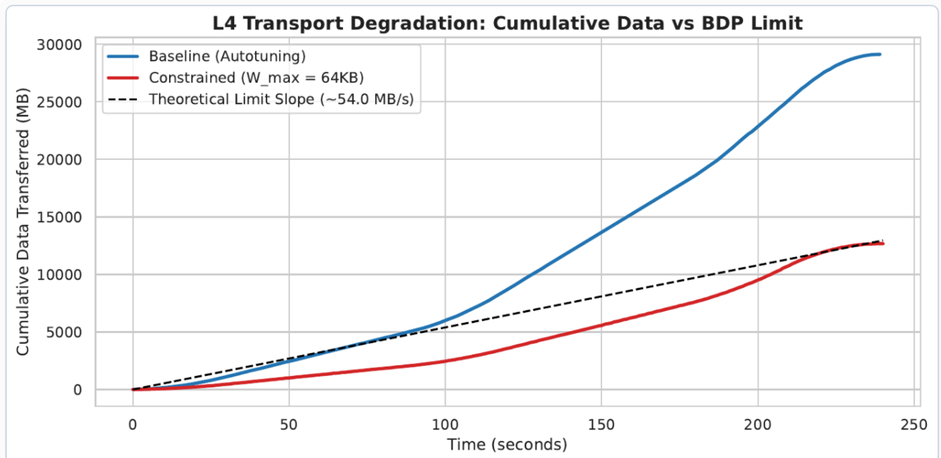


Fig. 9: Cumulative data transfer comparison. Under constrained TCP windows, the empirical transfer slope perfectly overlaps with the 54.0 MB/s theoretical BDP limit.

L4 Throughput Dynamics & TCP ACK Bundling

- **Dynamic Time-Series Dynamics:**
  Sustained average bounded to **~54.0 MB/s** vs. highly volatile, oscillating instantaneous throughput.
- **Transient Spikes (~100 MiB/s):**
  Extreme temporary throughput bursts driven by WAN jitter, intermediate queueing delays, and multi-socket concurrency.
- **TCP ACK Compression & Bundling:**
  WAN latency forces ACKs to accumulate in intermediate buffers, hitting the sender in rapid-fire clusters and releasing stalled windows.
- **Coordinated Sender Stalling:**
  Immediate post-burst depletion of the constrained **64 KB** window triggers synchronized socket stalls awaiting the next ACK cluster.

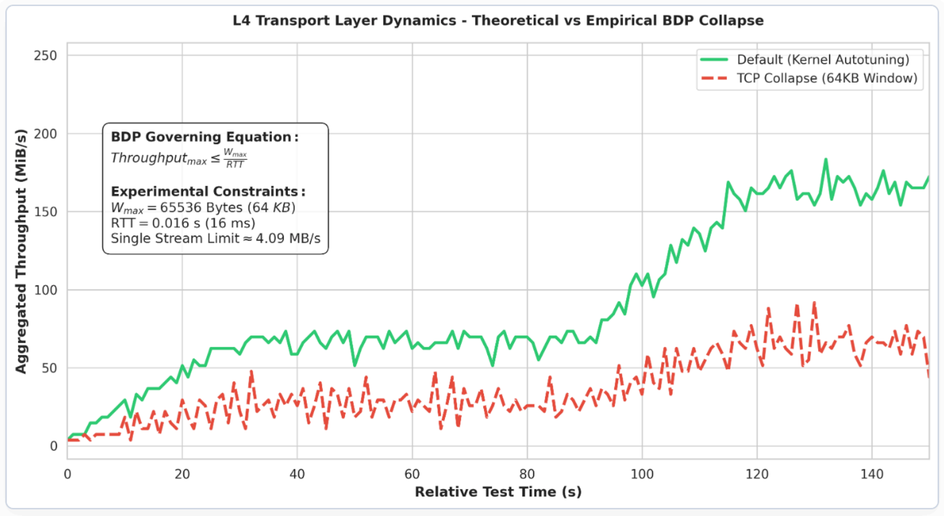


Fig. 10: Instantaneous L4 throughput dynamics. The volatile oscillations in the constrained regime validate the dynamic transition between ACK bundling bursts and sender stalling.

CAP Theorem Validation via Latency ECDFs

- **Sigmoidal Cumulative Distribution (ECDF):**
  The empirical latency profiles exhibit a mathematically canonical sigmoidal ($\int$) geometry. This exceptionally steep near-vertical transition indicates a near-zero variance, validating highly deterministic performance with negligible network jitter under stationary cloud continuum conditions.
- **The Consistency Penalty (ΔT):**
  A rigid, parallel horizontal displacement is observed between the two distributions:
  - **Eventual Consistency (AP / BASE):** Median latency of **72 ms** (optimized local core write at Frankfurt).
  - **Strong Consistency (CP / ACID):** Median latency of **88 ms** (synchronous blocking core + edge write at London).
- **Empirical Proof of Brewer's CAP Theorem:**
  The measured delta is exactly:

$$\Delta T = T_{CP} - T_{AP} = 16\ ms$$

  which correlates precisely with the physical WAN Round-Trip Time (RTT) between London and Frankfurt, demonstrating that synchronous consensus geographic blocks impose a non-negotiable physical transport penalty.

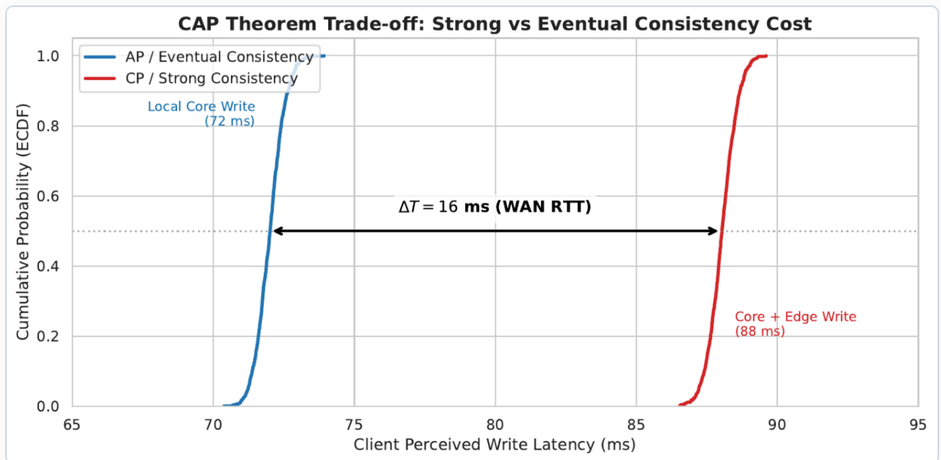


Fig. 11: Latency ECDF of CP vs. AP consistency models. The deterministic 16 ms horizontal shift provides empirical proof of the CAP theorem at Layer 7.

Storage Backend Trade-offs & The Throughput Paradox

- **The Throughput Paradox:**
  Under an intense concurrent workload of 500 VUs, S3-based cloud object storage (Spaces) running on the BASE paradigm achieves a **42% reduction** in client-perceived average latency (**101.15 ms**) compared to the highly optimized, local SSD transactional ACID path (**176.80 ms**).
- **The Physical Root (Eliminating Queue Starvation):**
  This paradox is driven by L7 offloading. Writing directly to local SSD forces all traffic synchronously through the core host and NGINX proxy, saturating the block controller and inducing severe kernel queuing delays ($\mathbf{T_q}$). Conversely, Harbor's HTTP 307 Redirect bypasses NGINX entirely, delegating concurrent bulk streaming directly to the highly parallel S3 distributed backend.
- **Converged Transport-Layer Throughput:**
  Despite S3's latency advantage, both storage backends converge to an identical aggregate throughput plateau of **≈36.80 MB/s**.
- **The Ultimate Leveler:**
  Once local host storage bottlenecks are bypassed, geodistributed data transfer remains strictly bounded by Layer 4 transport physics (WAN RTT and Bandwidth-Delay Product constraints).

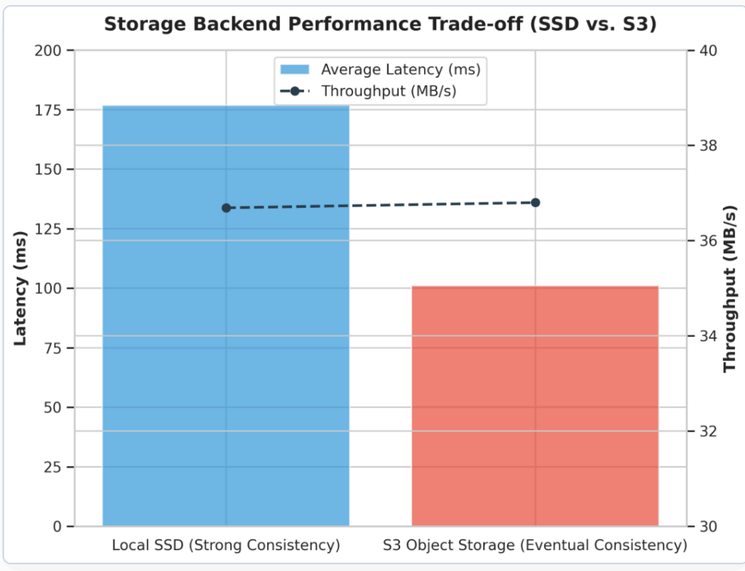


Fig. 12: Storage performance trade-off under peak concurrency (500 VUs), highlighting the convergence of aggregate throughput under geodistributed WAN limits.

System Resource Footprint & Scaling Profiles

- **The Static Resource Tax (m_i):**
  Harbor's modular architecture imposes a high static tax, requiring a baseline of **604 MB** of RAM at idle to maintain namespaces, cgroups, and loopback networks for 10+ internal containers.

- **VM Host vs. Container Isolation:**
  To resolve telemetric discrepancies, we distinguish the total VM host footprint (~1.5 GB under peak load, including OS kernel, NGINX proxy, and monitoring agents) from the strictly isolated middleware stack peaking at **1085 MB** (Harbor) vs. **843 MB** (Registry:2).

- **Sustained Compute Scaling Profiles (500 VUs):**
  - **Harbor (Compute-Bound / Self-Induced Thrashing):** Average CPU is low (**31.9%**) due to S3 data-plane offloading. However, peak stress drives CPU to 100% due to intense kernel scheduler thread contention and PostgreSQL connection starvation.
  - **Registry:2 (I/O-Bound / Steady State):** Locks the host CPU at a sustained **86.1%** (peaking at 100%) in a highly productive steady state, actively proxying and processing every network and disk block transaction.

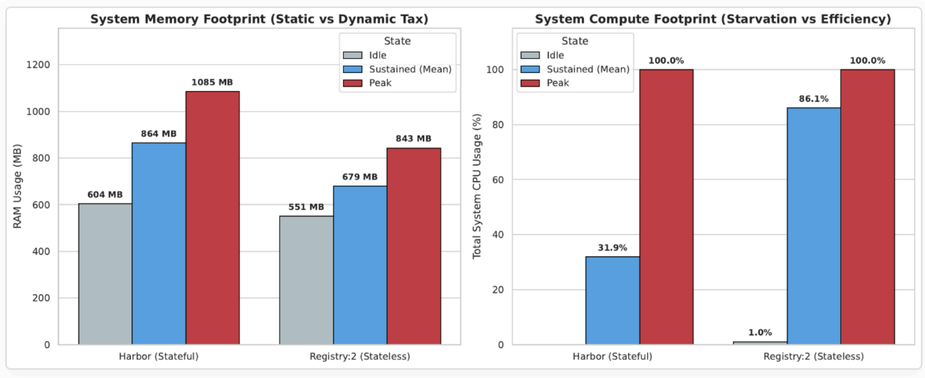


Fig. 13: Comparative resource footprint (RAM and CPU) under peak concurrency (500 VUs), contrasting stateless efficiency against stateful modular tax.

Appendix: Full-Stack Temporal CPU Characterization

- **Temporal Telemetry (~2800s):**
  Continuous 50-minute monitoring of 6 alternating payload stress-test cycles, contrasting NGINX proxy, Registry:2 (stateless), and Harbor (stateful) CPU profiles.
- **Multi-Core Compute Scaling:**
  Host Y-axis scales to 160%, capturing parallel multi-threaded execution across 2 vCPUs on the Frankfurt Core server.
- **Datapath Decoupling Proof:**
  NGINX CPU drops to **near 0%** during Harbor peaks. Post-authentication, the binary payload is routed to S3 via HTTP 307 Redirect, completely bypassing the local proxy.
- **Unified Path Overhead:**
  NGINX CPU spikes to **30%-50%** in perfect sync with Registry:2, driven by synchronous L7 proxying, buffer parsing, and context-switching.
- **Architectural Insulation:**
  Stateful control planes, despite their baseline tax, successfully insulate front-end proxies from L7 bufferbloat and CPU starvation under heavy geodistributed load.

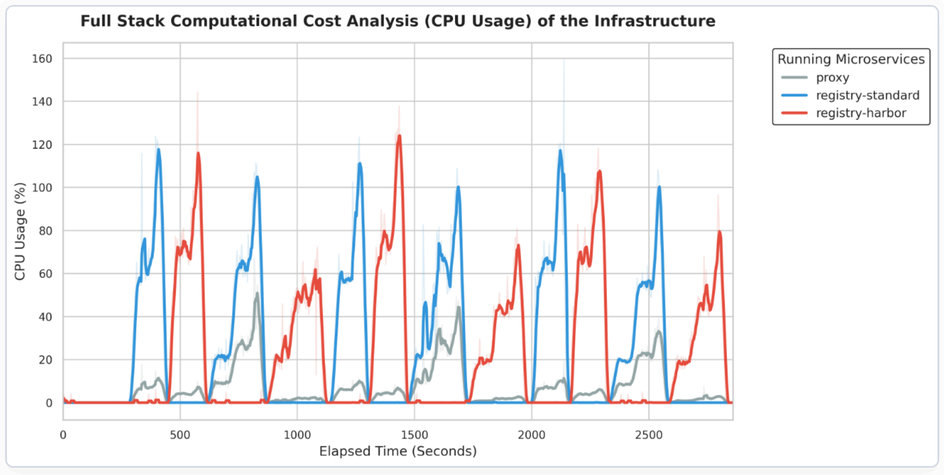


Fig. 14: Full-stack CPU utilization time-series over alternating heavy workloads, providing empirical proof of L7 datapath offloading.